



P A C K E T A D E

Midway Review

Six-Month Checkpoint – Flagship Deliverable

Full multi-fleet review of the PacketADE Agent Development Environment

7 domain deep dives · 22-agent bug hunt (16 confirmed, all fixed)

Competitive & technology briefs · 4-lens vision panel · 90-day plan

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1. Executive Summary

At its six-month midpoint, PacketADE is a real product, not a prototype. The engineering quality in its hardest subsystems — atomic state persistence with corruption recovery, a concurrency-correct agent turn lifecycle, battle-hardened xterm/PTY handling, a versioned sidecar protocol with restart supervision, and above-average SSH security — is well beyond what solo desktop projects usually achieve. Its breadth is its moat: an eight-provider chat matrix mixing subscription OAuth with API keys and local models, SSH-remote execution for terminals and agent tools, worktree-isolated parallel Flights with cost rollup, a memory layer, an MCP hub, and Windows-native support. No competitor bundles all of this.

Weaknesses cluster in three bands, none of them structural rot:

- **Lifecycle and recovery seams.** Both criticals and most highs lived where sessions cancel, error, crash, or restart — the seams the test suites never exercised against real providers.
- **Process infrastructure.** Zero CI, no coverage measurement, unsigned installers, no auto-updater, no crash telemetry, and macOS/Linux targets never compiled. The quality ladder is good — and entirely voluntary.
- **Daily-driver polish.** Ctrl+K stealing keystrokes from terminals, forgotten view on every launch, one-click session-destroying close, ~96 silent failure sites, a flagship launch modal that forgets its targets.

~177k	16 / 0	2 · 9 · 5	16 of 16	8	0
LOC under review	bugs confirmed / refuted	critical · high · medium	bugs fixed this session	chat providers, one contract	CI workflows / signed builds

Headline strengths

- Transport-agnostic api-agent:* event contract spanning the in-process runtime and the Node sidecar.
- Durable-by-design persistence: atomic writes, quarantine-first recovery, backend-owned merge semantics.
- SSH posture: host-key pinning, sh_quote everywhere, keyring passwords via askpass, fail-closed containment.
- Test depth where risk is highest: 165 Vitest files, 417 Rust tests, 14 sidecar smoke gates, a schema drift gate.
- Flights: one-prompt-N-agents with collision gating, Rust-enforced reviewer gates, per-flight cost rollup.

Headline risks (post-fix)

- Default "auto" mode maps to --dangerously-bypass-approvals-and-sandbox for Codex sessions (critical; open).
- No Flight recovery across restarts: zombie attempts, and failed attempts destroy their uncommitted work (open).
- Discipline-only quality ladder: at 177k LOC with no CI, one skipped gate ships regressions silently.
- Distribution stuck: unsigned, updater-less releases cap the user base at approximately one.
- Gated-write approvals hang forever on the flagship Claude subscription provider (pending_edit missing toolUsed — one-line fix, outside the 16-bug scope; open).

What was fixed this session

All 16 confirmed bugs repaired (15 distinct defects; two findings reconciled as one), with regression tests. Highlights: the Anthropic cancel-kills-conversation critical, Codex flight-cost over-counting critical, merge failures masquerading as success, the zombie-sidecar hang, session-bricking on advisory errors, and the localStorage migration ordering defect. Also: Gemini CLI removed (45 files), docs rewritten for honesty, statusline shipped in a sibling repo, marketing identity produced. Verdict: the project has earned a "ship it to more humans" posture — after the trust seams close, in the order of Section 8.

2. State of the Codebase

Seven domain deep dives. Each chapter: reviewer summary, selected strengths, ranked problems (severity-tagged), recommendations. Problems fixed in this session's bug loop are marked [FIXED]; everything else is open.

2.1 Frontend — React 19 / TypeScript / Zustand

A mature, unusually well-documented codebase (~119k LOC in src/) with sophisticated engineering in its hardest areas: the rAF-coalesced streaming pipeline with flush-before-done ordering; xterm/PTY handling working around an xterm 6.0 parser crash, WebGL blanking, and StrictMode double-spawns; and disciplined hydration with sessionGlue as the single eslint-enforced bridge between engines. 50+ store test suites; inline design rationale makes a 55-store architecture legible.

Weaknesses: state-management scale effects (one conversations array holds every transcript; always-mounted chrome recomputes O(all-messages) cost per streaming frame; unmemoized rows re-parse markdown per frame); persistence fragmented across five mechanisms; and dead code from rapid churn that now misleads (vestigial pane registry, routing card wired to nothing, persisted view never restored).

Strengths

- Stream coalescer: one store write per animation frame; tail chunks never drop or reorder.
- Battle-hardened xterm layer with sequence-aware transcript replay.
- Enforced engine decoupling via sessionGlue; strong hydration discipline.
- 50+ test suites incl. persistence-migration and 931-line listener tests.
- Code-splitting done right; live PTYs survive navigation.

Problems (ranked)

HIGH Prompt Library "Send to terminal" silently broken via dead layoutStore pane registry

(src/stores/layoutStore.ts)

addPane is never called; sendToTerminal always gets undefined — silent no-op. CLAUDE.md documents the dead path.

HIGH Always-mounted chrome subscribes to the full conversations array per streaming frame

(src/components/layout/LiveSpendChip.tsx)

Spend/agents chips and approval banner re-render at up to ~60 fps; cost aggregation re-runs over every message each frame.

MEDIUM Unmemoized message rows and MarkdownRenderer (src/components/agents/chat/MessageList.tsx)

Mount-once-never-unmount rows re-parse the entire conversation's markdown per frame after scrolling a long transcript.

MEDIUM Persistence fragmented across five mechanisms with hand-rolled duplication

(src/stores/layoutStore.ts)

setProjectPath re-implements workspaceStore persistence inline — a documented drift hazard.

MEDIUM routingStore task-routing UI is a dead end — resolveForTask has zero callers

(src/stores/routingStore.ts)

Users configure routing that has no effect anywhere; wire into launch paths or hide until LM1–LM7 lands.

MEDIUM Persisted selectedView written but never restored — always boots to Welcome (src/lib/bootstrap.ts)

Half the round-trip is dead code; daily users re-navigate every launch.

MEDIUM Monolithic view components with high hook density (src/components/views/GitHubView.tsx)

GitHubView 1204 lines / 23 hooks; WorkspaceCreationModal 1138 / 26; the hardest files to safely modify.

MEDIUM Fragile PTY approval automation and initial-prompt timing heuristics (src/hooks/useTerminalSession.ts)

Approvals write yln on regex TUI detection; non-Claude CLIs get a blind 3 s prompt delay that can swallow the task.

LOW Trailing-edge-only save debounce loses in-flight turns on crash

(src/stores/agentConversationPersistence.ts)

The 500 ms timer never fires during continuous streaming; add a max-interval flush.

LOW ~25 hardcoded "packetade:" literals despite brand.ts; dead files; 4 always-on status pollers; window CustomEvent side-channel (src/lib/brand.ts et al.)

A future rename is again multi-file churn; ServersView dead; 4 IPC polls per 5 s idle; untyped cross-component events.

Recommendations

- Memoize rows + renderer; derived chrome selectors; split conversation metadata from bodies.
- Delete the vestigial pane API; repoint promptStore; fix CLAUDE.md.
- Restore selectedView on boot; centralize storage keys with an eslint ban on raw prefixes.
- Replace the 3000 ms prompt delay with readiness detection; add max-interval save flush.

2.2 Rust Backend — commands, core, supervision, persistence

A genuinely mature Tauri v2 backend. storage.rs implements an atomic write path with a quarantine-first recovery ladder and hermetic tests; api_agent.rs has a correct turn-ownership protocol with cancellation at stream, tool-join, and pending-prompt levels; the supervisor rate-limits restarts and kills process trees on all platforms. Secrets in the OS keyring with careful migration. Production unwraps nearly all test-confined.

Weaknesses are architectural: one state.v1.json fully rewritten (pretty-printed, double-fsynced) on every slice save behind a spin-lock; and a trusted-webview security model — raw ssh argv from the frontend, a cosmetic PTY allowlist including shells, and PermissionMode::Auto running bash/write unprompted while MCP and web tools bypass the gate entirely.

Strengths

- Durable write path provably unable to clobber a good backup; crash-ordering tests.
- Concurrency-correct turn lifecycle with monotonic turn ids and deny-safe draining.
- Backend-owned-field merges prevent stale UI snapshots rewinding attempt state.
- Thoughtful SSH: SHA256 pinning, POSIX quoting, traversal rejection.
- Process hygiene: group kills, orphan reaping, crash fan-out, exit hooks.

Problems (ranked)

HIGH ssh_exec accepts arbitrary argv (and password) from the webview (src-tauri/src/commands/pty.rs:727)
ProxyCommand = local code execution; passwords in IPC. Build argv backend-side from ServerConfig id.

HIGH Default PermissionMode::Auto runs bash/write_file unprompted; gate covers two tool names
(src-tauri/src/commands/api_agent.rs:37)
MCP and web tools bypass the permission gate entirely. Default ask_for_risky; classify MCP/web.

MEDIUM Sync state writers spin on try_lock with 1 ms sleeps, no fairness (src-tauri/src/core/storage.rs:353)
Sync writers can starve indefinitely; bridge to fair async locking.

MEDIUM Monolithic state.v1.json rewritten on every slice save; memory slices unbounded
(src-tauri/src/core/storage.rs:100)
O(total-state) disk churn per UI persist; move to per-slice files or SQLite.

MEDIUM write_pty holds the global manager mutex across a blocking write (src-tauri/src/commands/pty.rs:615)
One wedged child freezes every terminal pane; per-session writer mutexes.

MEDIUM Usage JSONL append not serialized — interleaving can corrupt the cost log
(src-tauri/src/commands/usage.rs:35)
Single pre-joined buffer under a static Mutex.

MEDIUM PTY allowlist cosmetic; env injection unrestricted; in-process history unbounded
(src-tauri/src/commands/pty.rs:25, api_agent.rs:95)
Shells allowlisted and LD_PRELOAD/PATH passthrough; long sessions die on raw provider context errors.

LOW Docs contradict code on GitHub auth; legacy plaintext token path; unknown providers mislabeled as api-claude; timeout masking; 3 home-dir resolvers; 200 hand-registered commands (github.rs, api_agent.rs, lib.rs)
Migrate-and-delete the plaintext token; log unknown providers; adopt tauri-specta to end registration drift.

Recommendations

- Backend-constructed SSH invocations; ask_for_risky default; per-workspace trust toggle.
- Per-slice state files (interim) then SQLite; fair locking; manual restart_sidecar command.

- Context compaction for the in-process loop; typed error enums; tauri-specta.
- `deny(clippy::unwrap_used)` to lock in current discipline.

2.3 Sidecar & Provider Layer

The most architecturally interesting part of PacketADE: a versioned NDJSON stdio protocol (v11 — CLAUDE.md still said v10) connecting the Rust supervisor to a Node sidecar hosting three real providers, plus five in-process Rust providers sharing an OpenAI-compatible core. One event contract keeps the frontend transport-agnostic; the protocol file carries an inline changelog; the MCPH4 trust-snapshot design is genuinely well thought out.

Two systemic weaknesses: the protocol conflates advisory and terminal errors and the supervisor treated every error as session death (FIXED this session); and cancel/edit-approval correctness in the Anthropic provider — `cancel()` aborting the session-lifetime query (FIXED) and `pending_edit` omitting the mandatory `toolUseId` so gated write approvals can never resolve (STILL OPEN). The 14 smoke gates run against the echo provider — exactly the class of bug that shipped has no coverage.

Strengths

- Single api-agent:* contract across two backends; SSH-remote sidecars dropped in cleanly.
- MCPH4 trust snapshots: immutable authority, denial floors, Codex stdio trust proxy.
- Defensive I/O: 8 MB capped reader, UTF-8-safe truncation, SSE edge handling.
- Codex provider shows deep operational knowledge (shims, dual-schema parsing, watchdog).
- Thorough remote preflight; password-auth SSH refused on the protocol channel.

Problems (ranked)

CRITICAL [OPEN] Anthropic `pending_edit` omits `toolUseId` — gated write approvals can never resolve

(`agent-sidecar/src/providers/anthropic.ts`)

The resolver is parked under the id but the event carries none; the SDK stalls on the write tool forever. One-line fix.

HIGH [FIXED] Supervisor treats every error event as session-terminal (`src-tauri/.../agent_sidecar/handler.rs`)

Advisory errors permanently unrouted live sessions and leaked sidecar-side sessions and MCP children.

HIGH [FIXED] Anthropic `cancel()` aborts the long-lived query (v10 contract breach)

(`agent-sidecar/src/providers/anthropic.ts`)

Post-cancel sends pushed into a dead query and spun forever. See bug spotlight, Section 3.

HIGH [FIXED] Codex cumulative token snapshots additively accumulated into Flight cost

(`src-tauri/.../agent_sidecar/handler.rs`)

Sum-of-prefix-sums inflation. See bug spotlight, Section 3.

MEDIUM Cancel done/pump-done race; MCP trust false denials; no manual sidecar restart; Codex retry loses context; `allow_always` not sticky; no ack/nack; echo-only smoke gates (`agent-sidecar/*`)

A stale late done can zero a new turn; `get_dataset` denied for containing "set"; 3 crashes in 60 s costs all subscription chats until app restart; retry can replay a wake trigger; version skew degrades silently.

LOW `StartSessionRequest` god-object with undocumented aliases; v11 changelog drift; echo provider in production list; model-supplied regex in grep (`protocol.ts` / `mod.rs`)

The wire contract is wider than its type; documentation lockstep already broken; catastrophic backtracking can pin the shared event loop.

Recommendations

- Terminal/advisory distinction on the wire + regression test that ownership survives advisory errors.
- Ship `toolUseId` on `pending_edit` (one line) and a `restart_sidecar` command.
- Real-provider lifecycle gates behind env credentials: edit approval, cancel-continue, error-then-send.
- `node:test` runner; generate Rust protocol structs from `protocol.ts`; word-boundary trust heuristics.

2.4 Flights, Worktrees, SSH Remote Execution & the Remote Agents Plan

A genuinely mature one-prompt-N-agents orchestrator: Rust owns attempt lifecycle behind a serialized lock; the frontend re-hydrates from the backend as source of truth. Merge protection, dual-side path-claim collision gating, publish-before-teardown ordering, and Rust-enforced reviewer gates show hard-won correctness. SSH security is notably above average.

The weakest area is failure recovery across process death: startup recovery never touches attempts (zombie running attempts after a crash), Failed transitions force-delete uncommitted agent work, and SSH cleanup on event-driven failures never happens. The Remote Agents plan (Cloudflare relay, PWA-first, desktop-owns-execution) is thorough — and 100% unimplemented; Sprint 0 has not started and two gating decisions remain open.

Strengths

- Backend-authoritative state; terminal attempt states cannot be rewound by stale saves.
- Fail-closed host-key pinning for async launches; askpass-delivered passwords.
- Collision gating re-validated under the state lock; reviewer gate enforced in Rust.
- Cooperative-flight integration branches with conflict pausing and explicit landing.
- Hermetic money-path tests; partial-launch failure rehydration.

Problems (ranked)

CRITICAL [OPEN] No startup recovery for attempts — zombie running attempts after restart/crash

(src-tauri/src/core/orchestrator.rs)

Dead sessions, listeners on events that never fire, leaked worktrees. Resume plumbing exists and is unused.

HIGH [OPEN] Failed/rejected attempts destroy uncommitted agent work (src-tauri/src/core/worktree.rs)

worktree remove --force on every terminal transition; the evidence needed to diagnose failure is deleted.

HIGH [OPEN] SSH remote worktrees orphaned on event-driven failures

(src/stores/asyncAttemptTerminalListeners.ts)

Cleanup is deferred to a frontend call that never comes; the cancel path already demonstrates the backend fix.

MEDIUM Unbounded pkt/* branch and worktree accumulation; no GC (src-tauri/src/core/worktree.rs)

Dozens of stale branches within weeks; .pkt-worktrees/ pollutes git status.

MEDIUM Sequential multi-target provisioning; fingerprint never verified against known_hosts; attempt-level cost never populated; store persistence swallows errors (flight_attempts.rs / execution.rs / flightStore.ts)

4x serial SSH latency; pinning is a boolean; users cannot compare attempts on spend; silent write failures.

MEDIUM Remote Agents plan has zero implementation (dev/remoteargents/06-implementation-plan.md)

No relay, protocol package, PWA, or flag. High-quality plan; "planning package" is the honest status.

LOW Memory brief skipped for SSH targets; draft-PR target follows global repo selection; all-cancelled flights read "paused" (src/stores/asyncFlightStore.ts)

Remote attempts quietly behave worse; PRs can publish against the wrong repo.

Recommendations

- Attempt-aware startup recovery + orphan-worktree sweep — the biggest reliability gap.
- Stash/commit WIP before worktree removal; backend-owned SSH cleanup everywhere.
- Worktree/branch GC; per-attempt cost; first-class attempt diff review.
- Remote Agents Sprint 0 (protocol package, relay + PWA skeletons, flag) and resolve the two open decisions.

2.5 Security

Substantially better than typical for a solo-developed desktop agent tool: keyring everywhere with booleans-only across IPC, defensively engineered remote SSH, strict CSP, monitor-window invoke allowlist, no dangerouslySetInnerHTML. The most serious gap is permission-mode semantics: default "auto" maps to

--dangerously-bypass-approvals-and-sandbox for Codex while mapping to prompting modes elsewhere — three different safety postures behind one label, the most dangerous as default. Several IPC validations are defense-in-depth theater under the honest trusted-webview model; acceptable, but it should be stated.

Strengths

- API keys never cross IPC; keyring failures distinguished from missing credentials.
- Exemplary Unix askpass password path; POSIX-quoted, workspace-confined remote commands.
- Strict CSP; monitor windows restricted to a 5-command read-only allowlist.
- OAuth parsing extracts only expiry/refresh booleans; tokens never leave Rust.

Problems (ranked)

CRITICAL [OPEN] Default "auto" maps to --dangerously-bypass-approvals-and-sandbox for Codex

(agent-sidecar/src/providers/openai-codex.ts)

A stock subscription conversation runs unsandboxed and unapproved; prompt injection becomes silent host code execution.

HIGH Silent TOFU fallback when SSH host fingerprint is unpinned (src-tauri/src/core/execution.rs)

Legacy servers stay in accept-new forever with an invisible warning; badge the state; hard-require pinning for password hosts.

MEDIUM Windows stdin password piping can leak to the remote command; unused npx/node shell grants; unbounded plaintext PTY transcripts; raw ssh argv from the webview (tool_runtime_ssh.rs / capabilities / pty.rs)

The Unix-fixed hazard persists on the primary platform; ACE primitives granted for zero benefit; secrets-at-rest grow forever.

LOW Stale specs-gen.vercel.app CSP entries; MCP rawConfig ships Authorization headers to the webview; caller-supplied fs confinement; auto-install of CLIs on SSH connect; full "repo" scope device flow

(tauri.conf.json / mcp.rs / fs.rs / useServerConnection.ts)

Remove dead allowances; redact secrets pre-IPC; align editor fs with the agent runtime; make remote installs opt-in.

Recommendations

- Permission-mode unification with truthful labels — required before handing the app to anyone else.
- Visible SSH trust state; written threat model in dev/; transcript lifecycle management.
- Security-posture panel showing the effective flags actually passed per session.

2.6 Daily-Driver UX — shell, navigation, keyboard, lifecycle

Well-considered where ambient awareness matters: pinned approval banners, a running-agents-and-spend HUD, PTYs that survive navigation, one status-rollup truth. The unified Composer is the strongest single surface in the app.

What would grind an 8 h/day user down is the keyboard and lifecycle model: five scattered keydown listeners with inconsistent guards, layout-dependent chords, no pane cycling, a nav-only palette missing core destinations, one-click session-annihilating close, forced Welcome on boot, no undo, ~96 console-only failure sites vs one toast consumer, and a flagship launch modal that forgets its targets and ignores Escape. None of it is architectural — the primitives all exist.

Strengths

- PinnedApprovalBanner with one-click jump from any view.
- RunningAgentsChip + LiveSpendChip: a persistent fleet-and-spend HUD.
- One Composer for launch and chat with drafts, attachments, voice, queued sends.
- Flight-launch safety rails; atomic persisted-state writes keep data safe at rest.

Problems (ranked)

CRITICAL [OPEN] Global Ctrl+K/Escape handler fires inside terminals and inputs (src/App.tsx)

No editable-target guard, no xterm custom key handler: readline kill-line pops the palette over your shell.

CRITICAL [OPEN] Close kills all PTYs/agents/attempts with zero confirmation; restart lands on Welcome

(src/components/layout/TitleBar.tsx)

No onCloseRequested guard anywhere; the persisted view is written but never read.

HIGH ~96 console-only failure sites vs one toast consumer; launch modal forgets targets and ignores Escape; palette nav-only and missing Flights/Agents/Cost; no undo anywhere; no workspace/pane keyboard switching (multiple)

Silent data-loss discovery; a re-selection tax on the flagship flow dozens of times a day; "Clear all memory" one native confirm from wiping months of patterns; layout-dependent chords broken on non-US keyboards.

MEDIUM Notification gating inconsistent, OS toasts are dead ends; one-click Bypass-perms toggle; startup blocks on full conversation hydration (src/lib/notifications.ts et al.)

Attempt completions require a fully hidden window and bypass prefs; a misclick removes the human gate; cold starts scale with history size.

LOW "New" menu promises sessions it lacks; Settings/Tools naming collision

(src/components/layout/Toolbar.tsx)

Three related-but-different "tools" concepts coexist.

Recommendations

- Terminal-safe shortcut scoping; close-confirmation with live counts; view restore.
- Command palette v2 (fuzzy, actions, recents) and pane cycling.
- Toast-tier error surfacing; one confirm primitive; undo via soft-delete.
- Sticky launch defaults; Escape-to-close default; central shortcut registry + cheat sheet.

2.7 Quality & Infrastructure — tests, gates, CI, release, telemetry

Unusually strong at the unit level, unusually weak at the process level. 165 Vitest files (50 of 54 stores covered), 417 Rust tests concentrated where risk is highest, 13 smoke scripts written from real regressions, a Rust-to-TS schema drift gate, cross-compile bundle assertions. The bundling pipeline is the best code in this review area.

But every gate is voluntary: no CI (explicit decision), no hooks, a full gate long enough to skip under pressure. Coverage is never measured; E2E is 134 lines of web-mode smoke with all IPC mocked; macOS/Linux have never been compiled. No updater, no signing, version triplicated, zero field visibility for crashes.

Strengths

- Real store-level coverage incl. migration fixtures and lifecycle suites.
- Schema drift gate between Rust DTOs and TS bindings.
- SHA256-verified Node fetching; build-failing cross-compile assertions.
- Layered, honest release gating; local crash capture end-to-end.

Problems (ranked)**HIGH No CI of any kind — all gates voluntary at 177k LOC** (dev/local-quality-gates.md)

Session logs show multi-week fix loops landing between full gate runs; a dumb push workflow fixes the class.

HIGH macOS/Linux documented but never compiled; no updater, no signing (dev/multi-platform-build.md, dev/updater-setup.md)

The first non-Windows user is the integration test; every release is a manual SmartScreen-flagged reinstall — the updater runbook is complete, only ops work remains.

MEDIUM E2E is mocked-IPC web smoke; coverage never measured; prune-sidecar destroys the dev env; no clippy/fmt gate; zero field crash visibility (playwright.config.ts et al.)

PTY, sidecar handshake, and flight lifecycle have no automated end-to-end coverage; sidecar TS dev silently breaks after every production build.

LOW Bespoke sequential .mjs smoke chain; version triplicated; Playwright keys off a CI env that can never be set (agent-sidecar/test, scripts/release-gate.mjs)

node:test would fix the chain with zero new deps; a bump script beats a detection gate.

Recommendations

- Incremental CI: one 10-minute workflow now, compile matrix later; clippy + rustfmt now.
- Updater first, then signing; release automation as a script that becomes the CI job.
- Measure coverage before writing more tests; one golden-path native smoke test.
- Opt-in crash report submission; persistent frontend error log.

3. Bug Hunt & Fixes

Six dimension finders, dedup, then adversarial verification with fix designs. All 16 findings survived the verify stage (0 refuted) and all were fixed this session by four fixer agents with disjoint file territories and regression tests. Findings #13 and #15 describe the same TOCTOU defect from two dimensions and were reconciled during fixing — 15 distinct defects repaired.

#	Sev	Title / file	Failure scenario	Fix
1	CRITICAL	Anthropic cancel() aborts the session-lifetime AbortController anthropic.ts:1073	Stop mid-turn, then any follow-up spins forever; all later turns swallowed until app restart.	APPLIED
2	CRITICAL	Codex flight cost over-count: cumulative turn_summary snapshots re-accumulated handler.rs:546	Sum of prefix sums — ~10x cost/token inflation persisted onto Flights.	APPLIED
3	HIGH	packetcode-to-packetade migration runs after store hydration (ESM hoisting) src/main.tsx:6	Upgraders see a reset app; first-session writes permanently overwrite migrated data.	APPLIED
4	HIGH	flight-planner:cost-updated emitted with no frontend listener handler.rs:567	Autonomy budget hard-stop and cost UI run on stale totals; caps overrun by a full attempt's spend.	APPLIED
5	HIGH	UTF-8 char-boundary panic in error_classifier::extract_message error_classifier.rs:186	Long non-ASCII errors panic mid-critical-section; attempts stick in Running forever.	APPLIED
6	HIGH	Reader loop exits on oversized (>8 MB) stdout line, leaving a zombie sidecar supervisor.rs:929	Every sidecar conversation freezes mid-turn with no error, no restart, chip still Ready.	APPLIED
7	HIGH	Every sidecar error event treated as session-terminal handler.rs:600	One transient Codex exit permanently bricks the conversation and leaks the sidecar session.	APPLIED
8	HIGH	Crash-recovery contract broken: "please resend" can never re-create the session agentTaskStore.ts:701	After a sidecar crash every resend fails with "No active session" until app restart.	APPLIED
9	HIGH	Non-conflict merge failure during cooperative integration reported as success worktree.rs:874	Task marked done while nothing merged; Land Flight lands without the accepted work.	APPLIED
10	HIGH	land_local_integration_branch leaves the user's checkout mid-merge worktree.rs:970	Conflicting landings strand MERGE_HEAD and conflict markers in the real repo.	APPLIED
11	HIGH	Pricing table missing default models (claude-opus-4-8, gpt-5, MiniMax-M3) pricing.rs:106	Flight rollups and usage.jsonl record \$0 for the default models; guardrails never trip.	APPLIED
12	MEDIUM	done handler auto-sends queued messages even when the turn was cancelled apiAgentListeners.ts:360	Cancelling an attempt re-sends the queued follow-up 0 ms later.	APPLIED
13	MEDIUM	Concurrent resumeApiConversation installs duplicate listener sets agentTaskStore.ts:431	Chunks applied twice (doubled transcript), spurious failed state, permanent listener leak.	APPLIED
14	MEDIUM	issue-watcher listener in Monitor windows clobbers the issues slice issueStore.ts:774	A stale monitor copy overwrites shared localStorage — comments and links destroyed.	APPLIED
15	MEDIUM	ensureApiAgentListeners TOCTOU (reconciled with #13 as one defect) agentTaskStore.ts:431	Same root cause via the events dimension; single-flight install + resume guard fixes both.	APPLIED
16	MEDIUM	write_file_contents cannot create new files commands/mod.rs:68	Project Rules save always fails on repos missing AGENTS.md or CLAUDE.md.	APPLIED

3.1 Spotlight: the two criticals

Critical #1 — Cancel kills the conversation. On the flagship Anthropic (Subscription) provider, `cancel()` ran the SDK's turn-scoped `interrupt()` and then unconditionally aborted the single session-lifetime `AbortController` owning the long-lived query serving every turn. The pump exited and never restarted while every other layer believed the conversation was alive (protocol v10 makes cancel non-terminal; the registry and Rust both kept the session). The next send pushed into a dead iterable: no chunk, no done, no error — a spinner forever, and every later turn silently swallowed until app restart. **Fix:** cancel is now turn-scoped — parked permission/edit resolvers drained first, `interrupt()` success tracked, session-lifetime abort only as a loud-failing fallback that ends the prompt so follow-ups error instead of hanging.

Critical #2 — Codex flight costs inflated ~10x. The Codex provider emits `turn_summary` on every token-count event with session-cumulative totals (its own comment: "replace, not accumulate"), and the frontend HUD correctly replaces — but the Rust flight rollup priced each snapshot as a per-turn delta and added it. N snapshots $T_1 < \dots < T_N$ accumulated $T_1 + \dots + T_N$ instead of T_N — sum of prefix sums, compounded for sub-agent addresses, persisted into state and shown in the cost chip. **Fix:** provider-aware accounting — for snapshot-semantics providers the handler tracks the last cumulative snapshot per (session, address) and accumulates only the positive component-wise delta with monotonic-reset re-baselining; wire events and frontend unchanged.

Pattern worth naming: 10 of 16 confirmed bugs live in cancel/error/crash/restart seams — precisely the paths deterministic echo-provider gates cannot reach. The highest-leverage test investment is a small set of real-provider lifecycle gates (edit-approval round-trip, cancel-then-continue, error-then-send) behind env-gated credentials.

4. Competitive Landscape & Tech Trends

4.1 The mid-2026 agent-orchestration market

- **Cursor 3** (Apr 2026): Agents Window — up to 8 parallel agents per prompt in isolated worktrees or cloud/remote machines, outputting PRs; computer-use verification; self-hosted cloud agents; Automations; its own fast model (Composer 2).
- **Windsurf -> Devin Desktop** (Jun 2026): Agent Command Center (Kanban of every local + cloud agent), Spaces, multi-agent support via the open Agent Client Protocol (ACP); Devin Cloud from \$20/mo.
- **Conductor**: the category-defining Mac-only app — parallel worktree agents, strong diff/PR flow, free with BYO subscription; used at Linear, Vercel, Notion, Stripe. Limits: macOS-only, no remote execution.
- **Shakeout**: Terragon Labs shut down (Jan 2026, open-sourced); Bloop shut down (Apr 2026) leaving Vibe Kanban community-maintained. Cloud-orchestrator burn rates are fatal; orphaned users are up for grabs.
- **Claude Code**: Agent Teams with fleet guardrails (20-subagent cap, depth 3); Desktop redesigned around parallel agents with per-session worktrees; sandboxed in-app browser; Remote Control (outbound-only sync to mobile).
- **VS Code** is a multi-agent hub (Agent Sessions, native Claude/Codex, /delegate); **Zed + JetBrains** standardized on ACP with an agent registry — "LSP for agents" replacing bespoke PTY scraping.
- **OpenCode / Crush**: BYOK terminal-first engines PacketADE can wrap, not direct competitors.
- **Table stakes** per the 2026 category survey: session visibility, isolation, diff review, workspace organization, resumption. Differentiators: mobile oversight, container sandboxing, SSH/remote execution (only Mux), IDE-level controls.

What PacketADE has that rivals lack

The 8-provider matrix (subscription OAuth + API keys + local models); SSH-remote execution for PTY and agent tools with host-key pinning (only Mux is comparable); Flights above issues/sessions with Kanban, GitHub integration, and per-flight cost rollup (cross-provider cost analytics is absent in every competitor); memory + MCP client/provider hub + prompt library + dictation + deploy in one app; Windows/Linux-native via Tauri.

Strategic implications

- Ship the Remote Agents plan with urgency; the differentiated angle is "your own SSH boxes as the fleet" — self-hosted-first, no cloud bill, reusable in the Anunta enterprise pitch.
- Adopt ACP as a second agent transport mapped onto the api-agent:* contract — one integration buys Devin Local, Junie, Goose, Cline, Auggie.
- Make PRs the first-class output of attempts with inline line-comment diff review.
- Add an agent-verified-outcome step (browser/UI verification pass, "verified" badge).
- Attention routing before full mobile: blocked/needs-review rollup + notifications, then the thin PWA.
- Scheduled/webhook Flight launches as triggers into the existing pipeline — not a planner-FSM revival; optional container isolation for attempts.
- Positioning: "Conductor for Windows/Linux + remotes"; "Devin Desktop without the vendor lock or cloud spend"; target Terragon/Vibe Kanban refugees.

4.2 Technology trends and implications

- **Claude Agent SDK**: nested sub-agents (depth 3), fallback chains, per-agent cost attribution, scoped permissions, subagent streaming; Sonnet 5 with 1M context / 128k output.
- **MCP spec 2026-07-28** (largest revision since launch): stateless HTTP core, Extensions framework, Tasks extension (durable long-running work), MCP Apps (sandboxed-iframe server UIs), auth hardening (iss validation; DCR -> client metadata), simplified elicitation; official Registry targeted Q4 2026.

- **OpenAI:** Agents SDK gained native sandboxing (Python-first; TS lags); Responses API replacing Chat Completions; Codex deprecated --full-auto for permission profiles; app-server JSON-RPC is the documented long-term integration path.
- **Local models:** Ollama 0.24 (MLX, ollama launch); Qwen3-Coder family, Devstral Small 2, Kimi K2.6 are genuinely agent-capable — validating the per-task-class local routing plan (LM1-LM7).
- **Sandboxing is a defined market** (~5 isolation tiers); Anthropic ships srt (Seatbelt/bubblewrap + network proxy); Docker Sandboxes GA'd microVMs; unsandboxed agent tools are becoming unacceptable for enterprise.
- **Platform:** keep xterm.js (WebGL on, Canvas deprecated); Tauri v2 is the safe base (v3 is a draft milestone); the updater plugin is mature and tauri-action produces signed multi-platform installers.

Adopt: MCP 2026-07-28 (+ Tasks mapped onto Flights; MCP Apps spike), registry awareness, Claude SDK upgrades + Sonnet 5, OS-level sandboxing for agent tools, Responses API, Tauri v2 updater now, richer Ollama routing defaults.

Watch/migrate: Codex exec -> app-server; Agents SDK TS sandbox parity. Obsolete: MCP session-stateful assumptions, Chat Completions as primary surface, --full-auto, xterm Canvas renderer, unsandboxed tools, DCR OAuth, manual-only MCP onboarding.

5. Feature Program

5.1 Fix now — open items after this session's repairs

Sev	Item	Domain	Why now
CRITICAL	Codex "auto" -> dangerously-bypass mapping	Security	Default mode silently runs unsandboxed host execution.
CRITICAL	Anthropic pending_edit missing toolUseId	Sidecar	Gated write approvals hang forever; one-line fix.
CRITICAL	Attempt-aware startup recovery	Flights	Crash mid-flight leaves permanent zombie attempts.
CRITICAL	Ctrl+K guards + xterm key handler	UX	The palette pops over your shell on readline kill-line.
CRITICAL	Close-confirmation + selected-view restore	UX	One misclick kills every session; every boot forgets state.
HIGH	Preserve failed-attempt worktrees	Flights	Failures are undiagnosable — evidence force-deleted.
HIGH	Backend-owned SSH cleanup on failure paths	Flights	Remote hosts accumulate orphaned worktrees.
HIGH	restart_sidecar command + chip action	Sidecar	3 crashes in 60 s costs all subscription chats.
HIGH	Backend-constructed ssh argv (retire raw ssh_exec)	Security	ProxyCommand hole; passwords in IPC.
HIGH	Fix/remove Prompt Library send-to-terminal; wire or hide routing card; SSH TOFU badge	Frontend	Settings that silently do nothing erode trust.

5.2 Change — structural

- **Persistence:** per-slice state files toward SQLite; cap memory events; fair locking; failures as toasts.
- **Permission model:** one mode vocabulary across all three runtimes; ask_for_risky default; MCP/web tools gated; an "effective sandbox" field so the UI renders reality.
- **Streaming performance:** memoized rows/renderer, derived chrome selectors, metadata/body split, true transcript virtualization.
- **Contracts over convention:** tauri-specta; protocol structs generated from protocol.ts; typed error enums; terminal/advisory error distinction on the wire.
- **Process:** incremental CI; clippy + rustfmt; coverage measurement; node:test sidecar runner; one native smoke test; single-source version bump.

5.3 Add next — features the reviews rated "needed"

Workflow: view/workspace restore on boot; global search across conversations/transcripts/flights; palette v2 + pane cycling; sticky flight-launch defaults; Escape-to-close; undo via soft-delete; a notification center with click-to-jump; attempt diff review; per-attempt cost.

Platform & trust: auto-updater then signing; compile matrix; worktree GC; known-hosts UI; transcript lifecycle; context compaction for in-process providers; real-provider smoke gates; Remote Agents Sprint 0; written threat model; secret redaction pre-IPC; opt-in crash reports.

Nice-to-have (selected): crash-safe streaming persistence, palette flight relaunch, tile-split layouts, consolidated status poller, light theme, sticky allow_always, sidecar heartbeat, remote sidecar auto-deploy, per-flight repo binding, shortcut cheat sheet, cost sparkline, window-state restore, release automation script.

5.4 Daily-driver blockers — what would make you hate it

- **It fights your hands:** Ctrl+K in terminals; no pane/workspace keyboard switching; chords broken on non-US layouts; Escape ignored by the most-used modals.
- **It forgets:** Welcome-with-dead-panes every launch; flight targets re-picked every time; GitHub re-auth every morning.

- **It loses work silently:** unconfirmed close kills everything; crash loses the in-flight turn; restart zombies attempts; failed attempts delete their work; ~96 silent failure sites; "Clear all memory" one dialog from wiping months.
- **Its trust surfaces lied** [now fixed]: zombie cancels, bricking advisory errors, unusable crash recovery, ~10x Codex cost inflation, \$0 default-model pricing. [Still open]: edit-approval hang, sandbox-less default Codex mode, regex-guessed PTY approvals, silently unpinned legacy SSH hosts.
- **It litters and decays:** pkt/* branches accumulate forever; unbounded plaintext transcripts; whole-state rewrites become visible latency; long in-process chats die on raw context errors.
- **It's hard to keep and share:** manual SmartScreen-flagged updates; never-compiled macOS/Linux; invisible field crashes; auto-installing npm on every SSH host invites bans from controlled environments.

6. The Frontier — Vision Panel & the Top 12

Four lenses generated 38 raw ideas; a synthesis judge merged near-duplicates and ranked the twelve never-built features most worth doing.

6.1 Lens: Fleet Operations & Orchestration

Agent Reputation Ledger — Elo-style win rates from persisted attempt outcomes drive evidence-backed routing.

Flight Recorder — event + git-snapshot journal; scrub any attempt; fork-from-step with an injected correction.

Airspace Control — live cross-flight path/symbol claims; hold, reroute, or sequence before merge conflicts form.

Escalation Auction — capped-budget triage bids compete to win a failed attempt's relaunch.

Fleet Weather — provider health/rate-limit forecasting; ground-stop and divert queued work.

Shakedown Flights — replay your own completed flights as a private eval suite for new models/policies.

Filed Flight Plans — mine retrospectives into reusable templates with track records.

Tournament Mode — bracketed LLM judging over real worktrees with best-of composite assembly.

Convoy Corridors — typed dependency contracts between flights with auto hold/release and rebase.

6.2 Lens: Memory, Knowledge & Context

Flight-to-Runbook Distiller — compile successful flights into executable, parameterized runbooks.

Pitfall Tripwires — learned pitfalls gain triggers; the environment interjects warnings at runtime.

Agent Exit Debriefs — one structured question at session end, deduped into the pattern store.

Closed-Loop Context Compiler — token-budgeted briefs with usage-tracked automatic confidence rerating.

Memory A/B on Parallel Attempts — run the memory counterfactual for free on sibling attempts; measure ROI.

Nightly Consolidation Daemon — merge, decay, and promote patterns into AGENTS.md as reviewable diffs.

Decision Provenance — "why does this line exist" walks blame -> edit -> conversation -> pattern.

Memory Packs & Marketplace — signed, versioned bundles of patterns/runbooks installable like MCP servers.

Cross-Host Memory Mesh — learned patterns sync across the SSH fleet keyed by project.

6.3 Lens: Remote, Mobile & Collaboration

Unified Approval Inbox — risk-ranked triage queue across all hosts and Flights; batch low-risk approvals.

Time-Boxed Autonomy Grants — scoped expiring grants ("writes under src/ for 30 min") with visible countdown.

Agent Presence & Live Handoff — supervision as a transferable seat between desktop and phone.

SSH Fleet Mesh — migrate a running attempt to another capable host when the laptop lid closes.

Morning Brief — signed overnight digest per flight with swipeable merge triage.

Pair Tower — invite a second human with role-scoped approvals and per-human audit.

Dead-Man Escalation Chains — SLA-driven escalation for unanswered approvals; blocked-time metered.

Walkie-Talkie Mode — voice-native supervision with TTS narration and voice-answerable approvals.

Flight Radar — ambient wallboard of every attempt: phase, burn rate, blocked, staleness.

Guest Approval Links — lend one decision via a scoped, expiring capability URL.

6.4 Lens: Trust, Autonomy & Verification

Autonomy Licenses — agents earn expanded YOLO bounds through verifiable track record; breaches demote.

Signed Commit Provenance — attestation trailers linking commit -> attempt -> conversation -> evidence.

Adversarial Verification Lane — a red-team agent must break the diff or concede.

Blast-Radius Budgets — metered write scope per attempt — damage surface, not just spend.

Rollback Contracts — autonomy authorized only after the undo is generated and proven clean.

Taint Counterfactual Replay — re-run tainted turns with evidence redacted; divergence proves steering (moonshot).

Escrowed Claims — "tests pass" held in escrow until a deterministic verifier confirms it.

Policy Flight Recorder — deterministic replay of every autonomy decision; backtest policy changes.

Risk-Priced Verification Router — score each change and route to a proportional review lane.

Honeytoken Tripwires — decoy credentials trip instant hard stops on read/transmit.

6.5 The Top 12 — synthesized, ranked, sequenced

#01 Earned Trust Ledger: Reputation-Weighted Routing & Autonomy Licenses [difficulty: medium]

An append-only local ledger of per-attempt outcomes keyed by (agent x task type x repo area x model), computed from review verdicts, failure categories, cost-per-accepted-diff, and refuted claims — data already persisted but discarded. Drives evidence-backed routing and graduated autonomy tiers; the launch modal shows the receipt.

Sequencing: Build the data-capture layer FIRST — every week of delay is lost training data. Capture + read-only stats panel first; routing and tiers once the record has weight. It is the flywheel hub.

#02 Tournament Mode: Bracketed Judging, Best-of Assembly & Bid-Based Recovery [difficulty: medium]

A reviewer-agent bracket runs pairwise diff comparisons over parallel attempts, ranks a leaderboard, and can assemble a best-of composite in the integration branch; on failure the same machinery runs cheap diagnosis bids. All judgments feed the ledger. No other tool has worktrees + reviewer runtime + integration branch.

Sequencing: After ledger capture and a stable reviewer gate; the highest-visible-payoff first-wave item — it attacks eyeballing N diffs.

#03 Unified Approval Inbox with Leash Slider & Dead-Man Escalation [difficulty: medium]

Every permission/edit/plan approval across all hosts and Flights in one risk-ranked queue; scoped expiring autonomy grants as signed policies with countdown; SLA escalation with blocked-time metered next to cost.

Sequencing: Sequence with the Remote Agents plan — this IS the mobile MVP done right; ship the inbox desktop-first before the relay lands.

#04 Rollback Contracts: Pre-Proven Undo Insurance for Autonomy [difficulty: large]

Autonomous integration only after the revert is generated, dry-run-applied, and proven clean; a permanent one-click "unwind everything this policy did." Verified reversibility as a precondition of autonomy — "YOLO, but fully collateralized."

Sequencing: Build immediately after the auto-integration adapter ships, as its gating precondition; scope v1 to fast-forward-only integration.

#05 Flight Recorder: Time-Travel Replay & Counterfactual Forks [difficulty: large]

Journal every event plus git snapshots at tool boundaries; scrub any attempt with diff and reasoning in sync; fork-from-step into a fresh worktree with an injected correction. The same journal enables policy backtesting.

Sequencing: Start the journaling substrate early (record now, exploit later); defer scrubber and fork UX to a second wave.

#06 Runbook Distiller: Filed Flight Plans from Successful Flights [difficulty: large]

Compile a completed flight into an executable, parameterized template — task graph, routing, prompts, review criteria, budget, verification — offered at launch with its track record attached and versioned by later flights.

Sequencing: Mid-sequence: needs the ledger and a few dozen completed flights; ship distillation-on-completion first.

#07 Risk-Priced Verification: Blast-Radius Budgets + Proportional Review Routing [difficulty: medium]

Per-attempt quotas on files/dirs/LOC/manifests debited live; a risk score (radius, taint, path criticality, license tier) routes each change to a proportional lane, shown contestably on the AttemptTile.

Sequencing: Budgets first (standalone value when YOLO ships), router second; the connective tissue between the trust items.

#08 Closed-Loop Context Compiler with Exit Debriefs & Sleep-Cycle Consolidation [difficulty: large]

Token-budgeted briefs whose items are usage-tracked into automatic confidence rerating; a one-question exit debrief improves intake; a nightly daemon merges, decays, and promotes patterns into AGENTS.md as reviewable diffs. Memory becomes measured, garbage-collected, compounding.

Sequencing: Ship the two small pieces first (debriefs: days; daemon: contained), then usage tracking, then close the loop.

#09 Adversarial Verification Lane (Red-Team Gate) [difficulty: large]

An institutionalized attacker in an isolated worktree: "produce a failing test, crashing input, or injection escalation — or concede." Breaks auto-package as reviewer findings and ding the builder's license.

Sequencing: After Risk-Priced Verification — the adversary only makes economic sense as the high-risk escalation lane.

#10 Fleet Weather: Provider Health Forecasting & Ground Stops [difficulty: medium]

Aggregate rate-limit events, failure categories, and OAuth expiry into a per-provider weather map; ground-stop queued tasks, divert to the healthiest capable agent, hold for quota resets.

Sequencing: After the ledger's routing consumer — weather is the real-time modifier on its historical prior.

#11 Provenance Blame: Signed Agent Authorship + "Why Does This Line Exist" [difficulty: large]

Signed attestation trailers (keyring key, git notes) linking commit -> attempt -> conversation -> evidence, verifiable offline; a blame action walks to the conversation excerpt and the motivating pattern. The compliance story anchoring the enterprise pitch.

Sequencing: Record linkage metadata early; defer signing and blame UI until the trust-provenance loop lands; pull forward if the enterprise pitch goes live.

#12 Runtime Tripwires: Honeytokens + Active Pitfall Guards [difficulty: small]

Decoy credentials tracked by envelope ID trip instant hard stops, pause the Flight, demote the license, and write an incident record; learned pitfalls gain runtime triggers that interject warnings.

Sequencing: Ship the honeypot half FIRST in absolute sequence — the smallest item, an active intrusion alarm before autonomy expansion raises the stakes. (Cut from the 12: Flight Radar, Walkie-Talkie, Pair Tower, Guest Links, SSH migration, memory mesh, marketplace, convoy corridors, taint replay, presence/handoff.)

7. Actions Taken This Session & Verification

7.1 Fleets run

Research fleet — 14 agents: 7 domain deep dives, 2 web-research briefs, 4-lens vision panel + synthesis judge. **Bug-hunt fleet — 22 agents:** 6 dimension finders -> dedup -> adversarial verify + fix design; 16 confirmed (2 critical, 9 high, 5 medium), 0 refuted; findings #13/#15 reconciled as one defect — 15 distinct defects repaired.

7.2 Code and content shipped (uncommitted at report time)

- **Gemini CLI removed entirely:** 5 files deleted, 40 edited across TS + Rust; persisted "gemini" panes remap to plain terminal; retired agent ids dropped on hydration; the OpenRouter google/gemini-2.5-pro chat model intentionally kept. Both compile gates passed immediately after removal.
- **All 16 confirmed bugs fixed** with regression tests: turn-scoped Anthropic cancel; provider-aware Codex cost deltas; merge machine fails closed (local + SSH) and failed landings abort cleanly; zombie-sidecar tree-kill -> restart; per-turn errors no longer brick sessions; cost event renamed and wired to a listener; migration runs before hydration (ESM side-effect module); crash-recovery resend actually resumes; cancelled turns don't auto-send queues; single-flight listener installs; Monitor windows can't clobber issues; UTF-8 truncation panic fixed; pricing table gained claude-opus-4-8 / gpt-5 family / MiniMax-M3; write_file_contents creates new files with confinement preserved.
- **Statusline** (sibling repo claude-code-tools, committed 84f794f + pushed): Claude Code wiring verified byte-identical; research proved Codex supports only built-in status item ids and OpenCode has none — no fake adapters; a real tmux fallback segment shipped; docs corrected to verified item-id vocabulary.
- **Docs overhaul:** docs/index.html rewritten (fabricated stats, "Missions", Gemini, deleted-planner claims removed; honest status strip); roadmap.html rebuilt on the real R0-R11 tracks; the manual rewritten task-oriented; deferred-work.md re-verified; 14 stale dev/ docs archived; root README/ROADMAP/backlog/HANDOFF/CHANGELOG refreshed preserving the user's uncommitted edits.
- **Marketing:** hero poster, one-pager PDF, social banner, design-philosophy document ("Phosphor Deck").

7.3 Verification gates

Status recorded at report finalization by the orchestrator.

[PASS] pnpm build (tsc + vite, Windows toolchain)

vite built in 33.6 s, zero TS errors.

[PASS] cargo check (Windows toolchain)

only pre-existing ts-rs serde-attribute warnings.

[PASS] pnpm sidecar:build (sidecar tsc)

plus tsc --noEmit clean for app and sidecar.

[PASS] vitest run (src/stores/__tests__ + src/lib/__tests__)

1,260 tests across 166 files (440 Rust tests also pass via cargo test). Two pre-existing load-induced flaky tests were stabilized test-side during the session (timeout budget for cold module imports); every substantive failure found during gating was fixed.

[COMPLETE] Peer-review fleet over the full diff

4 reviewers (Gemini removal, Rust fixes, frontend fixes, docs truth): 3x approve-with-nits, 1x needs-changes; all 12 findings remediated — including one major (order-safe Codex cost delta accounting via per-session sequence stamping) — and re-verified green.

[QUEUED] Windows exe compile

runs immediately after this report is committed; artifacts and hashes will be recorded in HANDOFF.md.

7.4 Notable truths surfaced

- Zero CI, zero coverage measurement, unsigned installers, no updater, no crash telemetry — the quality ladder is discipline-only.

- `PROTOCOL_VERSION` is 11; `CLAUDE.md` said 10 (stale; gitignored/generated — verify against code).
- The "1,261 passing Vitest tests" figure came from a docs-agent count; verify against the vitest gate output before quoting externally.
- Deliberately left for the next loop: Prompt Library send-to-terminal, the no-op routing card, and the never-restored persisted view.

8. Recommended Next 90 Days

The data supports a clear order: lock in what was fixed, close the trust seams, unblock distribution, then start the frontier bets whose value compounds with time. Do not invert this — every frontier item assumes a foundation that does not silently lose work, and every week without CI risks re-breaking the 15 defects just repaired.

Phase 1 — Lock in & close the trust seams (days 0-14)

- **Green the gates and commit.** Run the six verification gates, land the diff, tag it. Nothing proceeds on an unverified base.
- **CI first — the cheapest insurance in this report.** One push-triggered workflow: pnpm lint, vitest run, cargo test. Add clippy + rustfmt while the ladder is open.
- **Close the four open trust items:** Codex "auto" -> workspace-write (bypass only explicit + warning chip); toolUsed on pending_edit; restart_sidecar command; stash-before-remove on failed worktrees.
- **Kill the top three rage-inducers:** terminal-safe shortcut guards, close-confirmation with live counts, selected-view restore.
- **Truth pass:** fix or hide the Prompt Library action and routing card; update CLAUDE.md (v11, GitHub auth, worktree paths, dead conventions).

Phase 2 — Reliability & distribution (days 15-45)

- **Flight recovery:** attempt-aware startup recovery with resume, backend-owned SSH cleanup, worktree/branch GC with auto-exclude — the single biggest reliability gap.
- **Updater, then signing.** The runbook made every decision (~30 lines of code); follow with an OV cert or Azure Trusted Signing; add the compile-only ubuntu/macos CI matrix in the same window.
- **Streaming performance pass:** memoized rows/renderer, derived chrome selectors, max-interval save flush.
- **Failure visibility:** route logSwallowed and persistence failures through the ToastProvider; one confirm primitive; opt-in crash reports.
- **Flights polish:** sticky launch defaults, per-attempt cost, attempt diff review, notification unification with click-to-jump.
- **Security batch:** backend-constructed ssh argv, prune npx/node grants and the stale CSP entry, SSH trust-state badges, transcript retention + 0600.

Phase 3 — Frontier bets & the remote flagship (days 46-90)

- **Start the two record-now-exploit-later substrates immediately:** the Earned Trust Ledger capture layer + stats panel (#1) and the Flight Recorder journal (#5) — their value compounds with every week of data.
- **Ship Runtime Tripwires' honeypot half (#12)** — the smallest Top 12 item, an active intrusion alarm before autonomy expansion.
- **Remote Agents Sprint 0 folded with the Approval Inbox (#3):** resolve the two blocking decisions (auth provider, encryption timing); protocol package + relay and PWA skeletons + feature flag; inbox desktop-first. The market brief is unambiguous — and "your own SSH boxes as the fleet" is the differentiated, Anunta-compatible angle.
- **Tournament Mode (#2)** once ledger capture exists — the highest-visible-payoff frontier item.
- **Standards spikes, timeboxed:** MCP 2026-07-28 (Tasks mapped onto Flights), an ACP transport spike on the api-agent:* contract, an OS-sandbox spike (srt/Landlock; Windows options); watch Codex app-server; adopt the Responses API.
- **Defer deliberately:** Rollback Contracts until auto-integration ships; the Adversarial Lane until risk routing exists; SQLite until per-slice files prove insufficient; full mobile until the inbox proves the loop.

The through-line: PacketADE's differentiators — provider breadth, SSH fleets, cost truth, and soon the trust economy no competitor is building — all depend on the same foundation: sessions that survive their own lifecycle, releases that update themselves, and gates that run without willpower. Ninety days of this sequence turns a formidable solo tool into something that can be handed to a second person without apology — precisely the bar the Anunta opportunity and any public beta will demand.